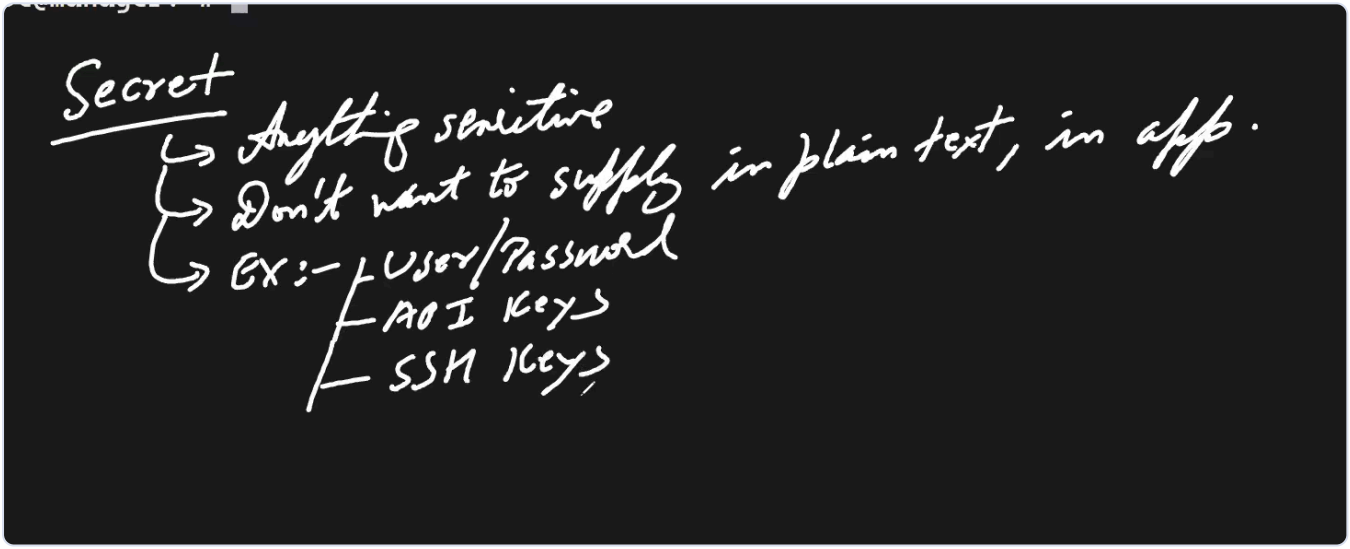
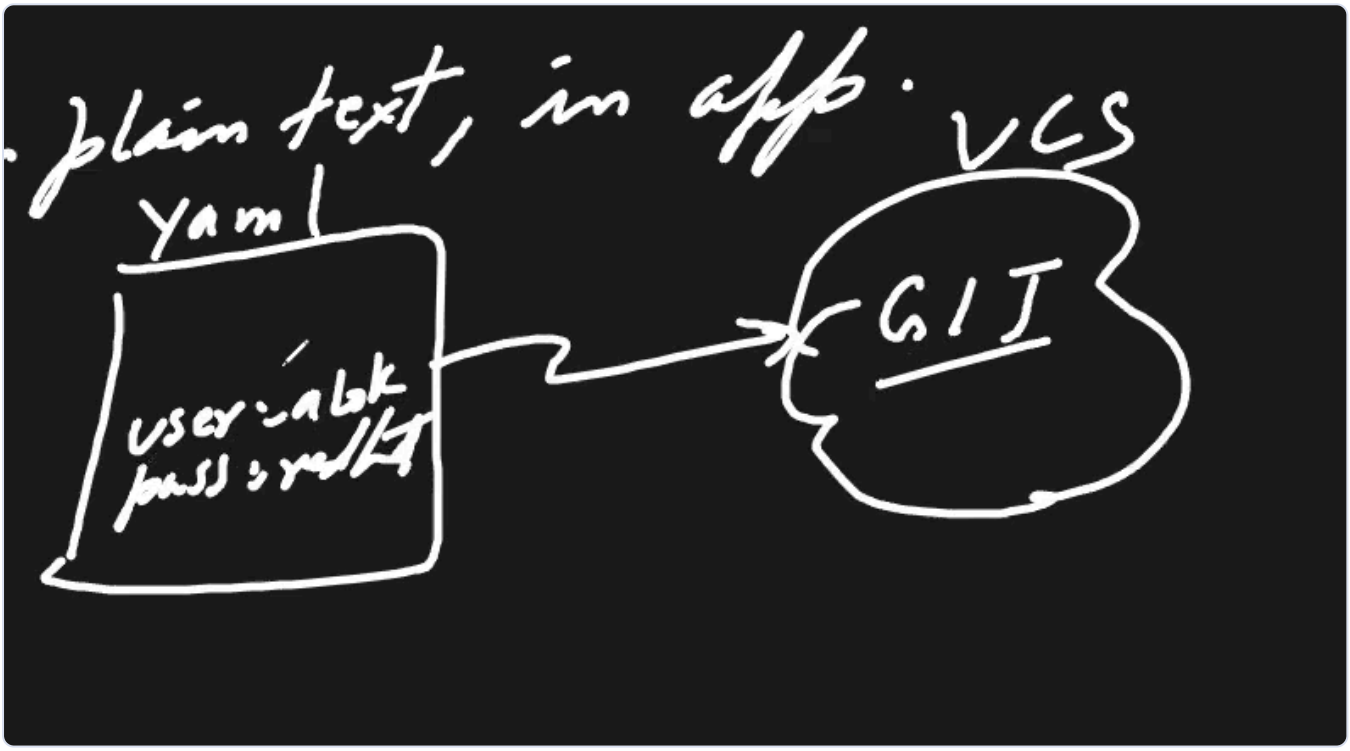
KUBERNETES PREPARATION · MY LEARNING NOTES · SESSION 08

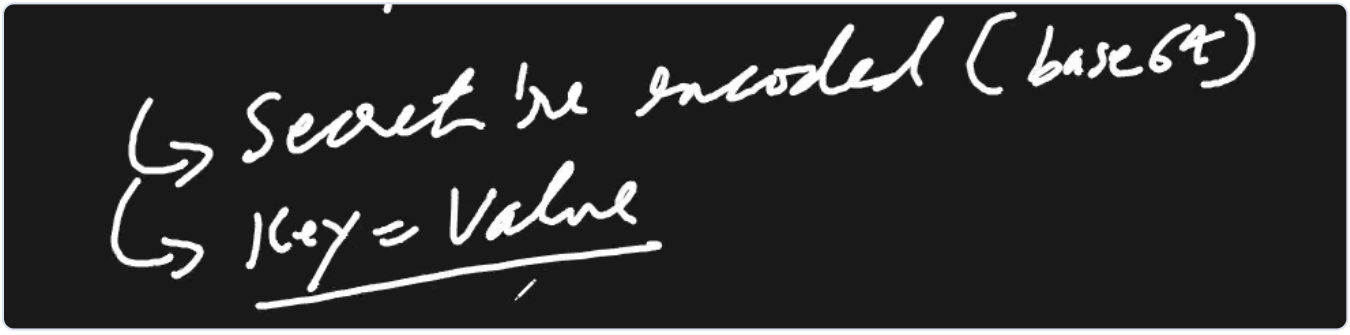
Secrets and ConfigMaps

Screenshot gallery — 45 captures, in session order


Secret
Anything sensitive
Don't want to supply in plain text, in app.
Ex:- User/Password
API Keys
SSH Keys




plain text, in app.
VCS
Yaml
GIT


↳ Secrets are encoded (base64)
↳ key = Value

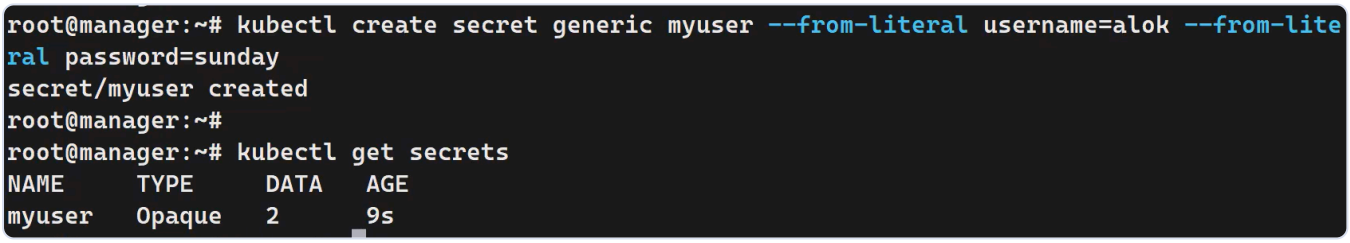

① Create secret
② Inject into Pod!

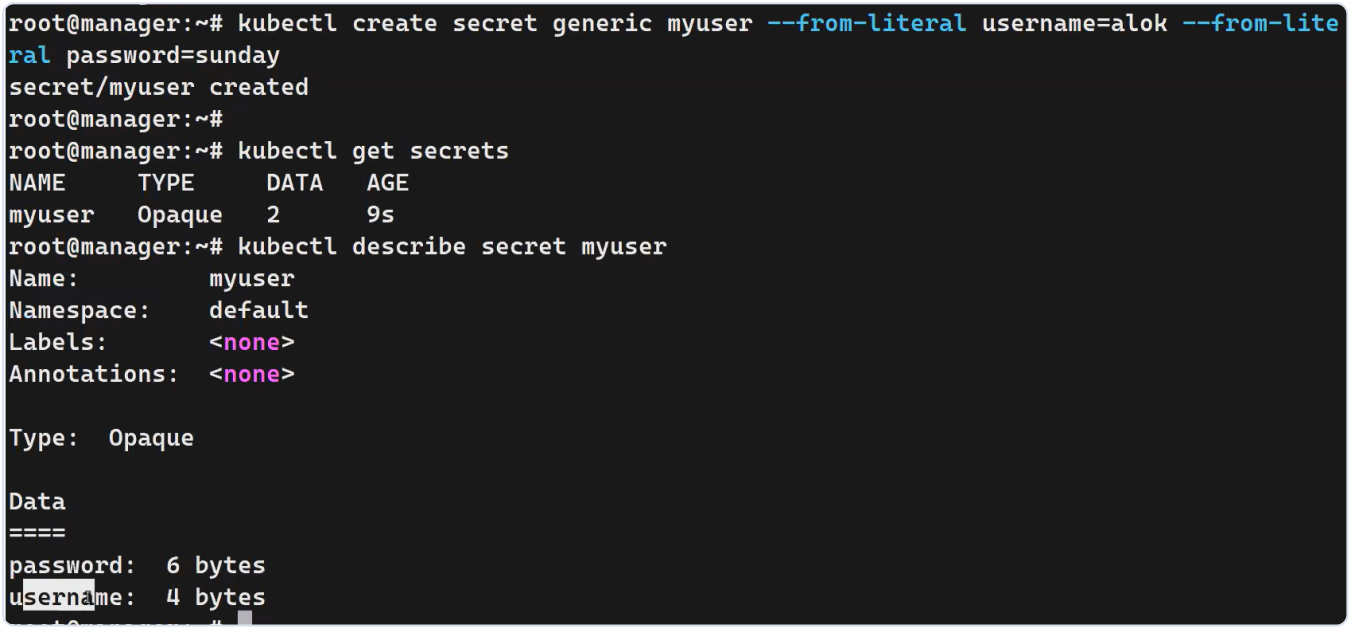
```
  # Create a new secret named my-secret with key1=supersecret and key2=topsecret
  kubectl create secret generic my-secret --from-literal=key1=supersecret
--from-literal=key2=topsecret

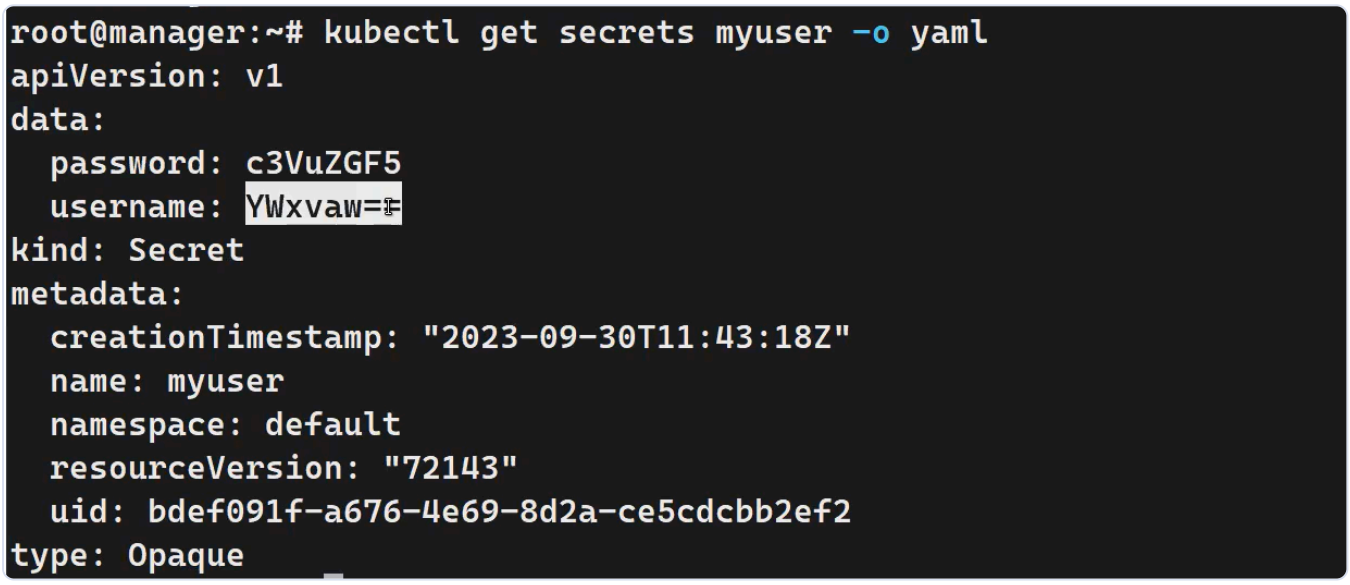
  # Create a new secret named my-secret using a combination of a file and a
literal
  kubectl create secret generic my-secret
--from-file=ssh-privatekey=path/to/id_rsa --from-literal=passphrase=topsecret

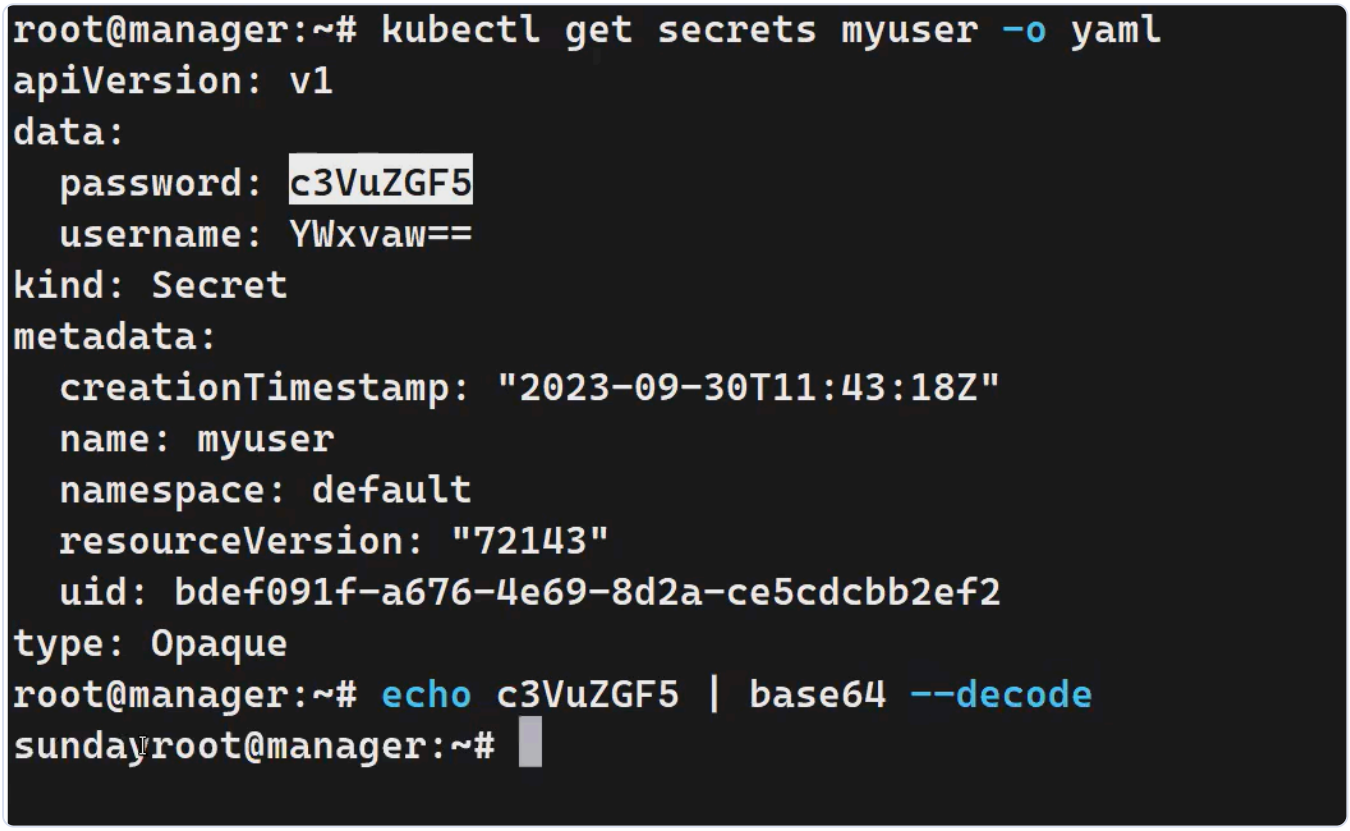
  # Create a new secret named my-secret from env files
  kubectl create secret generic my-secret --from-env-file=path/to/foo.env
--from-env-file=path/to/bar.env
```

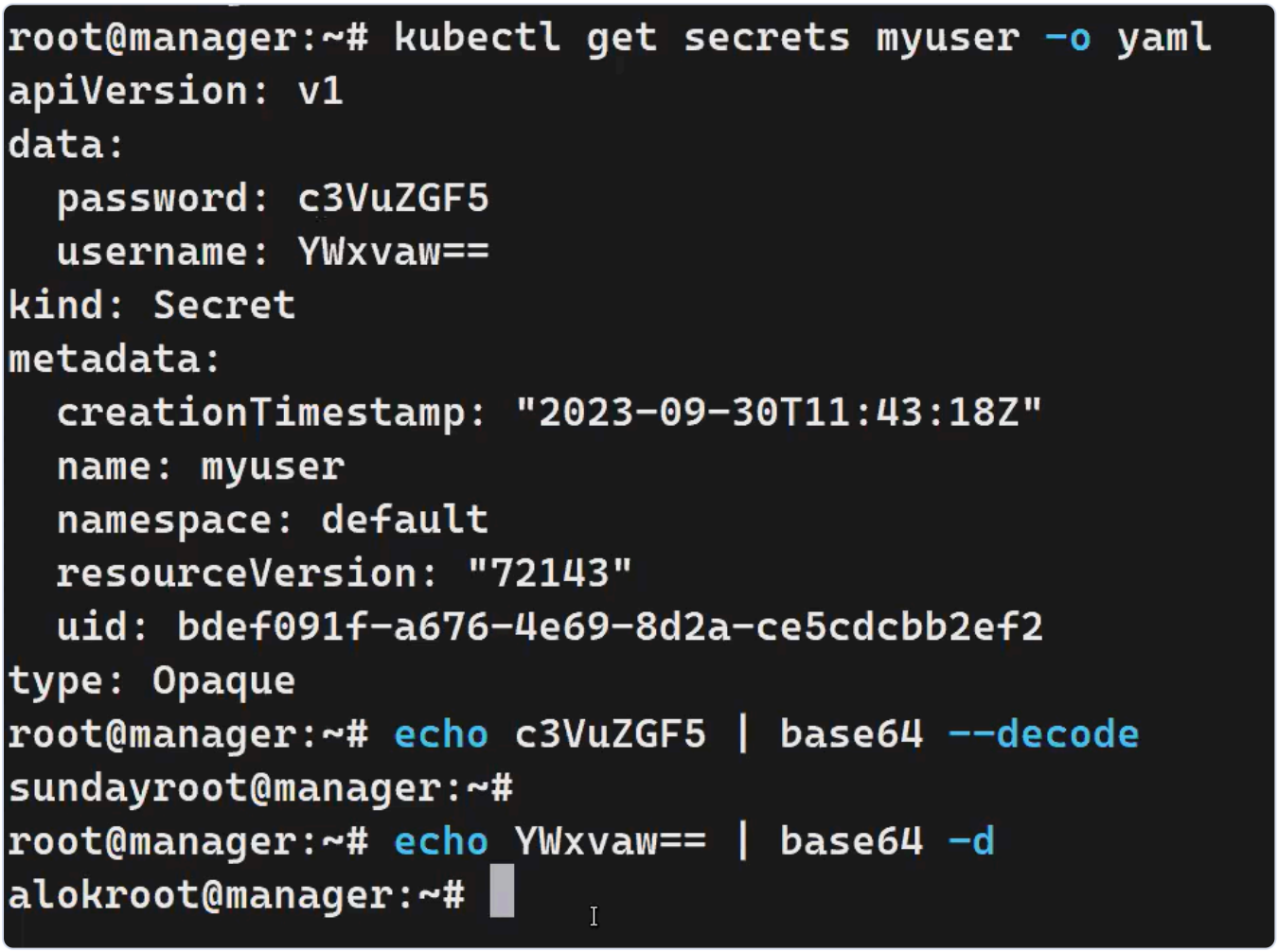
```
root@manager:~# kubectl create secret generic myuser --from-literal username=alok --from-lite
ral password=sunday
secret/myuser created
```

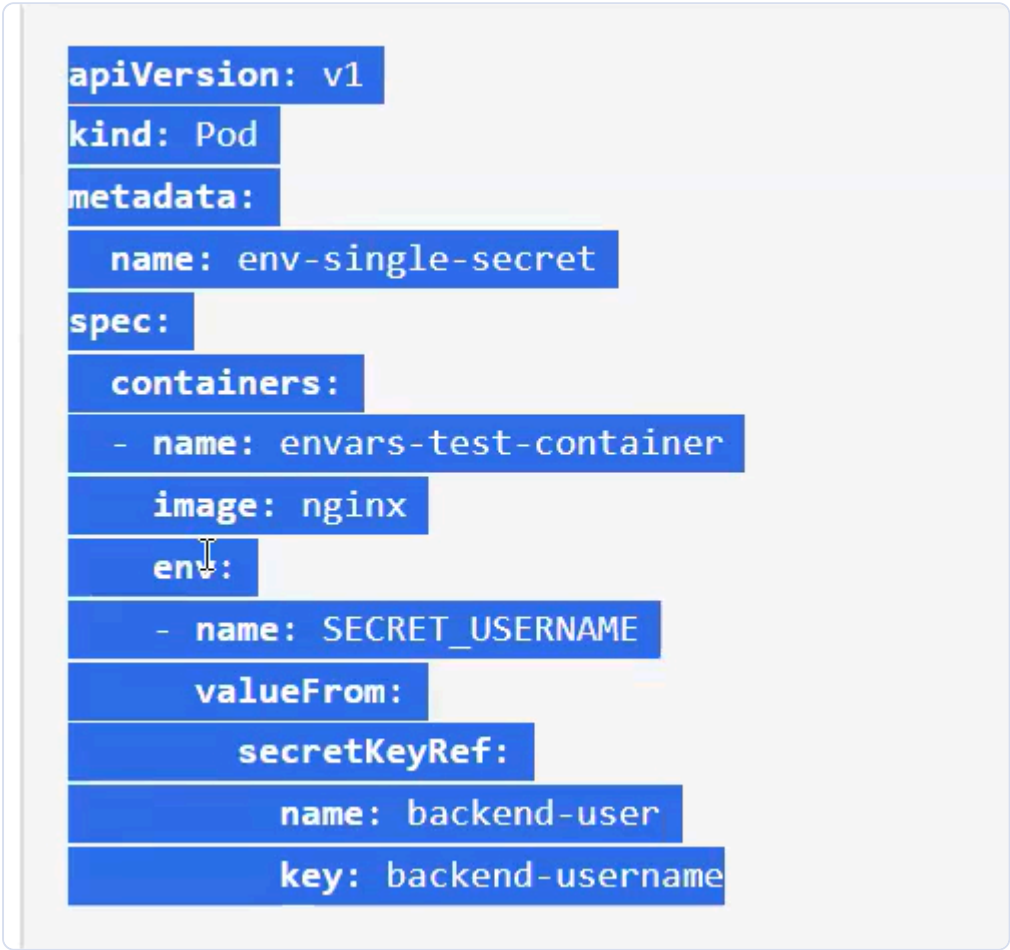

root@manager:~# kubectl create secret generic myuser --from-literal username=alok --from-lite
ral password=sunday
secret/myuser created
root@manager:~#
root@manager:~# kubectl get secrets
NAME TYPE DATA AGE
myuser Opaque 2 9s


root@manager:~# kubectl create secret generic myuser --from-literal username=alok --from-literal password=sunday
secret/myuser created
root@manager:~#
root@manager:~# kubectl get secrets
NAME   TYPE   DATA   AGE
myuser   Opaque   2   9s
root@manager:~# kubectl describe secret myuser
Name: myuser
Namespace: default
Labels: <none>
Annotations: <none>
Type: Opaque
Data
====
password: 6 bytes
username: 4 bytes


root@manager:~# kubectl get secrets myuser -o yaml
apiVersion: v1
data:
password: c3VuZGF5
username: YWxvaw=
kind: Secret
metadata:
creationTimestamp: "2023-09-30T11:43:18Z"
name: myuser
namespace: default
resourceVersion: "72143"
uid: bdef091f-a676-4e69-8d2a-ce5cdcbb2ef2
type: Opaque

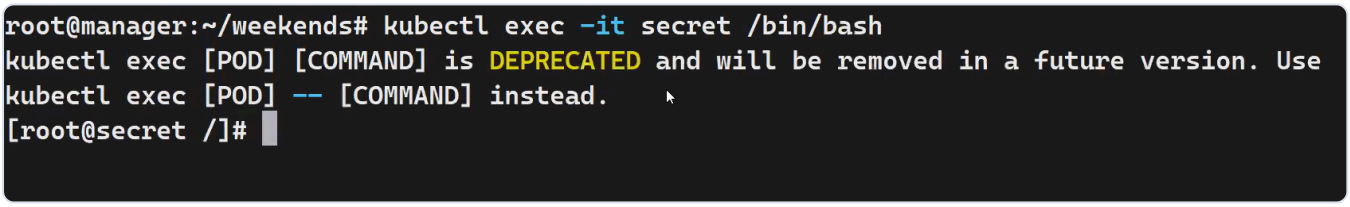

root@manager:~# kubectl get secrets myuser -o yaml
apiVersion: v1
data:
password: c3VuZGF5
username: YWxvaw==
kind: Secret
metadata:
creationTimestamp: "2023-09-30T11:43:18Z"
name: myuser
namespace: default
resourceVersion: "72143"
uid: bdef091f-a676-4e69-8d2a-ce5cdcbb2ef2
type: Opaque
root@manager:~# echo c3VuZGF5 | base64 --decode
sundayroot@manager:~#

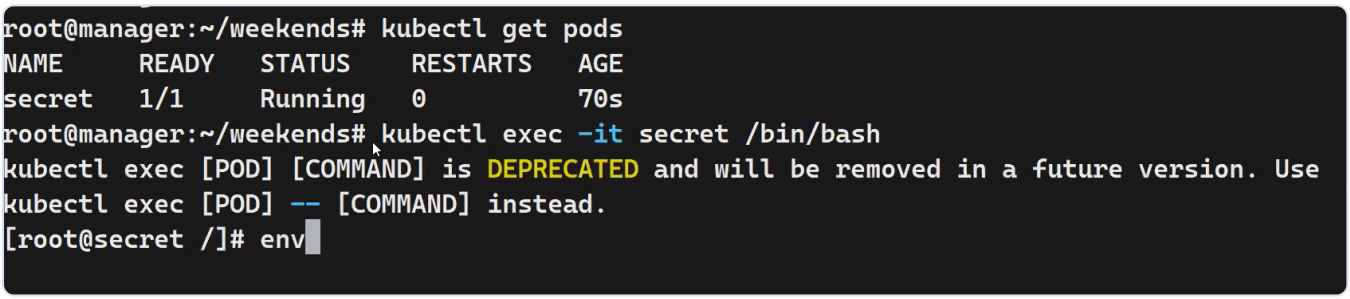

root@manager:~# kubectl get secrets myuser -o yaml
apiVersion: v1
data:
password: c3VuZGF5
username: YWxvaw==
kind: Secret
metadata:
creationTimestamp: "2023-09-30T11:43:18Z"
name: myuser
namespace: default
resourceVersion: "72143"
uid: bdef091f-a676-4e69-8d2a-ce5cdcbb2ef2
type: Opaque
root@manager:~# echo c3VuZGF5 | base64 --decode
sundayroot@manager:~#
root@manager:~# echo YWxvaw== | base64 -d
alokroot@manager:~#

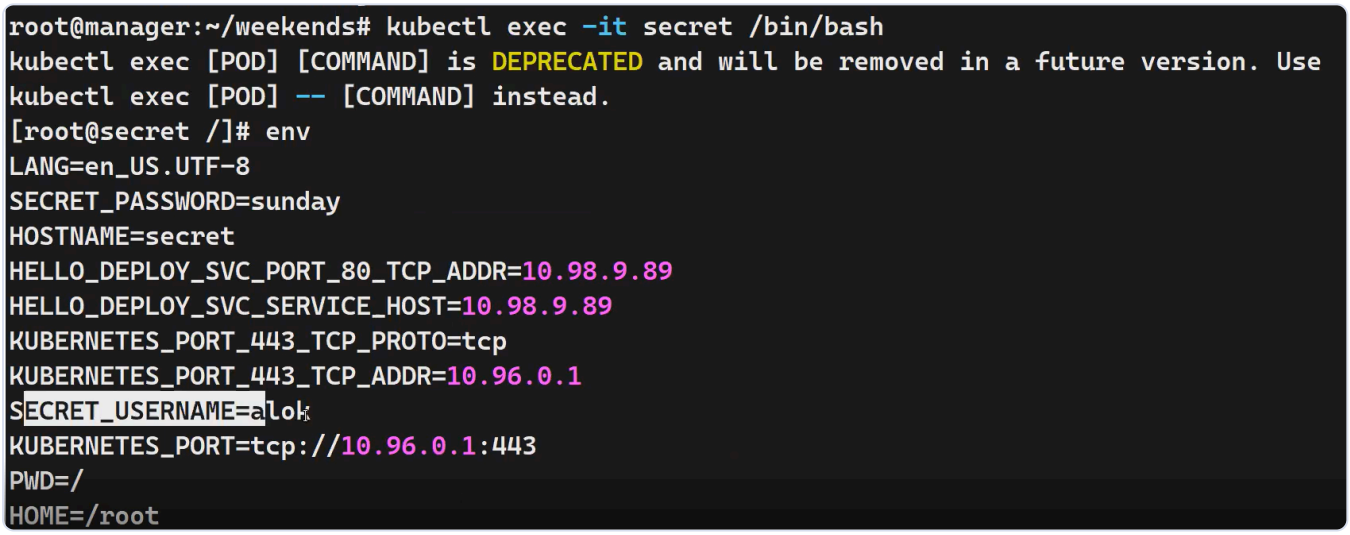

apiVersion: v1
kind: Pod
metadata:
name: env-single-secret
spec:
containers:
- name: envars-test-container
image: nginx
env:
- name: SECRET_USERNAME
valueFrom:
secretKeyRef:
name: backend-user
key: backend-username

```yaml
metadata:
  name: secret
spec:
  containers:
  - name: web
    image: lovelearnlinux/webserver:v1
    env:
    - name: SECRET_USERNAME
      valueFrom:
        secretKeyRef:
          name: myuser
          key: username
    - name: SECRET_PASSWORD
      valueFrom:
        secretKeyRef:
          name: myuser
          key: password
```

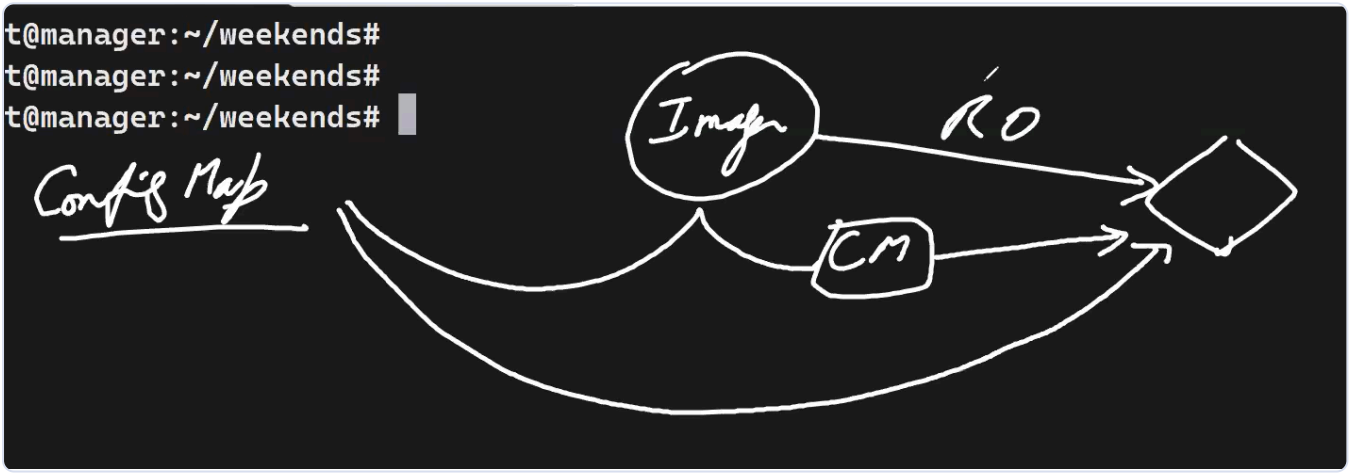

Environment:
SECRET_USERNAME: <set to the key 'username' in secret 'myuser'> Optional: false
SECRET_PASSWORD: <set to the key 'password' in secret 'myuser'> Optional: false

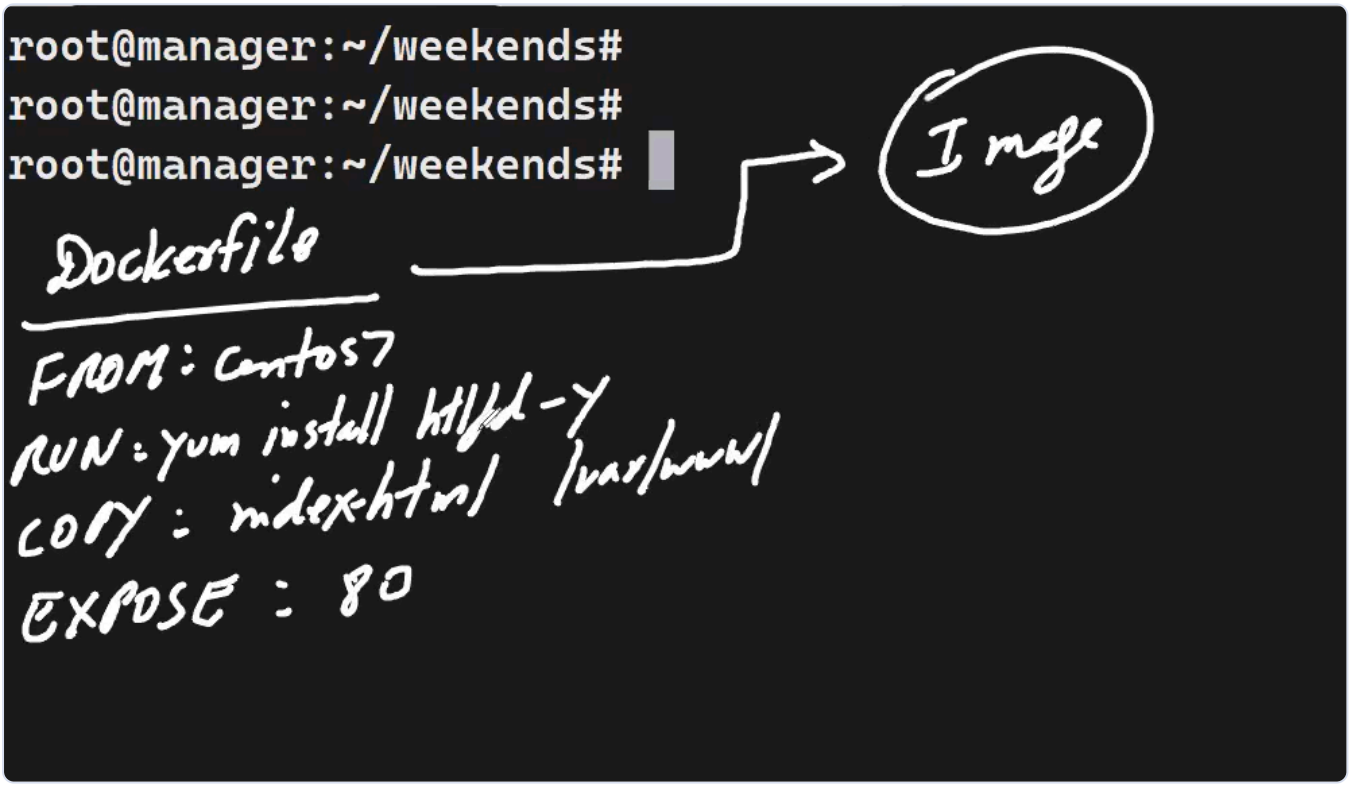

root@manager:~/weekends# kubectl exec -it secret /bin/bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use
kubectl exec [POD] -- [COMMAND] instead.
[root@secret /]#

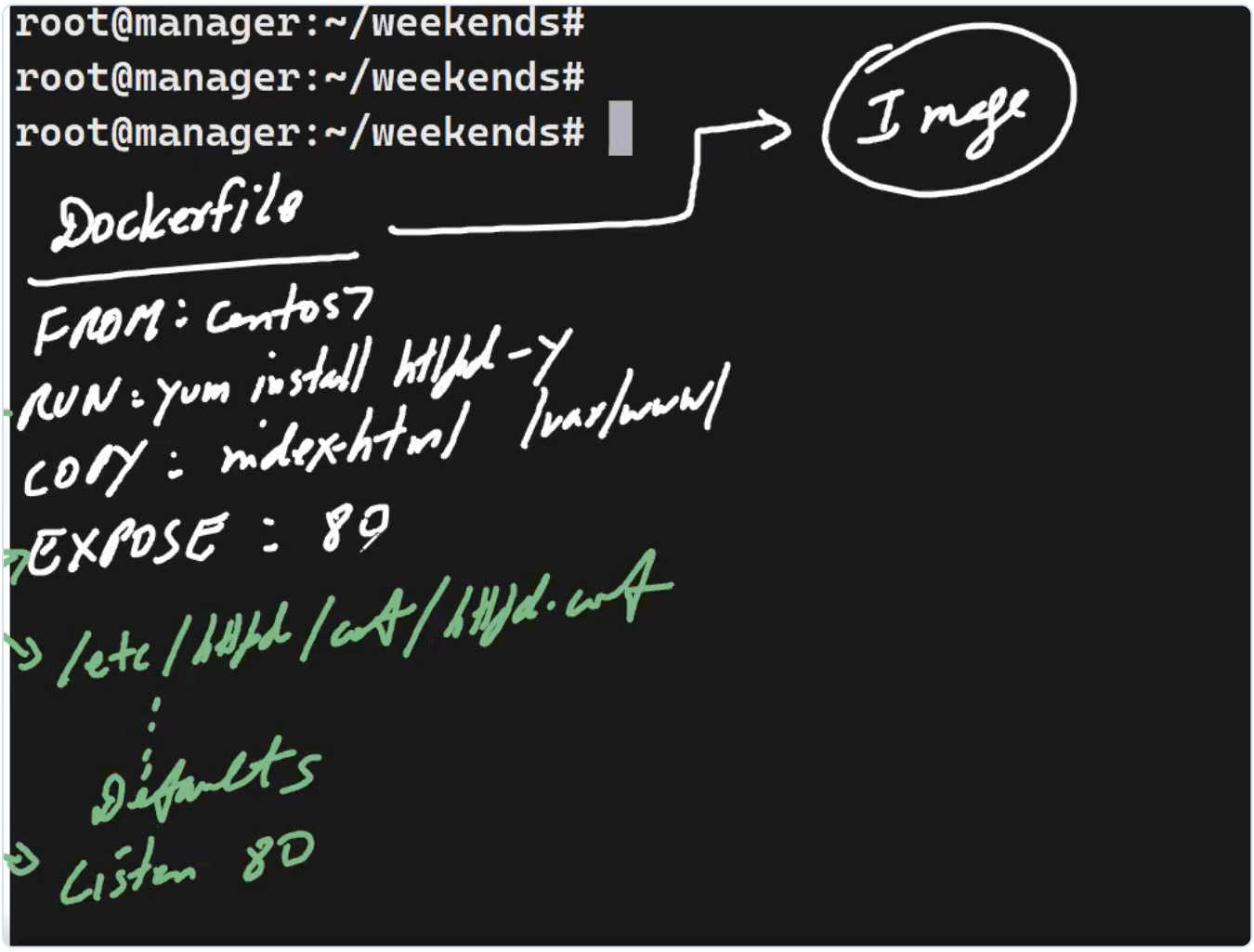

root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
secret 1/1 Running 0 70s
root@manager:~/weekends# kubectl exec -it secret /bin/bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use
kubectl exec [POD] -- [COMMAND] instead.
[root@secret /]# env

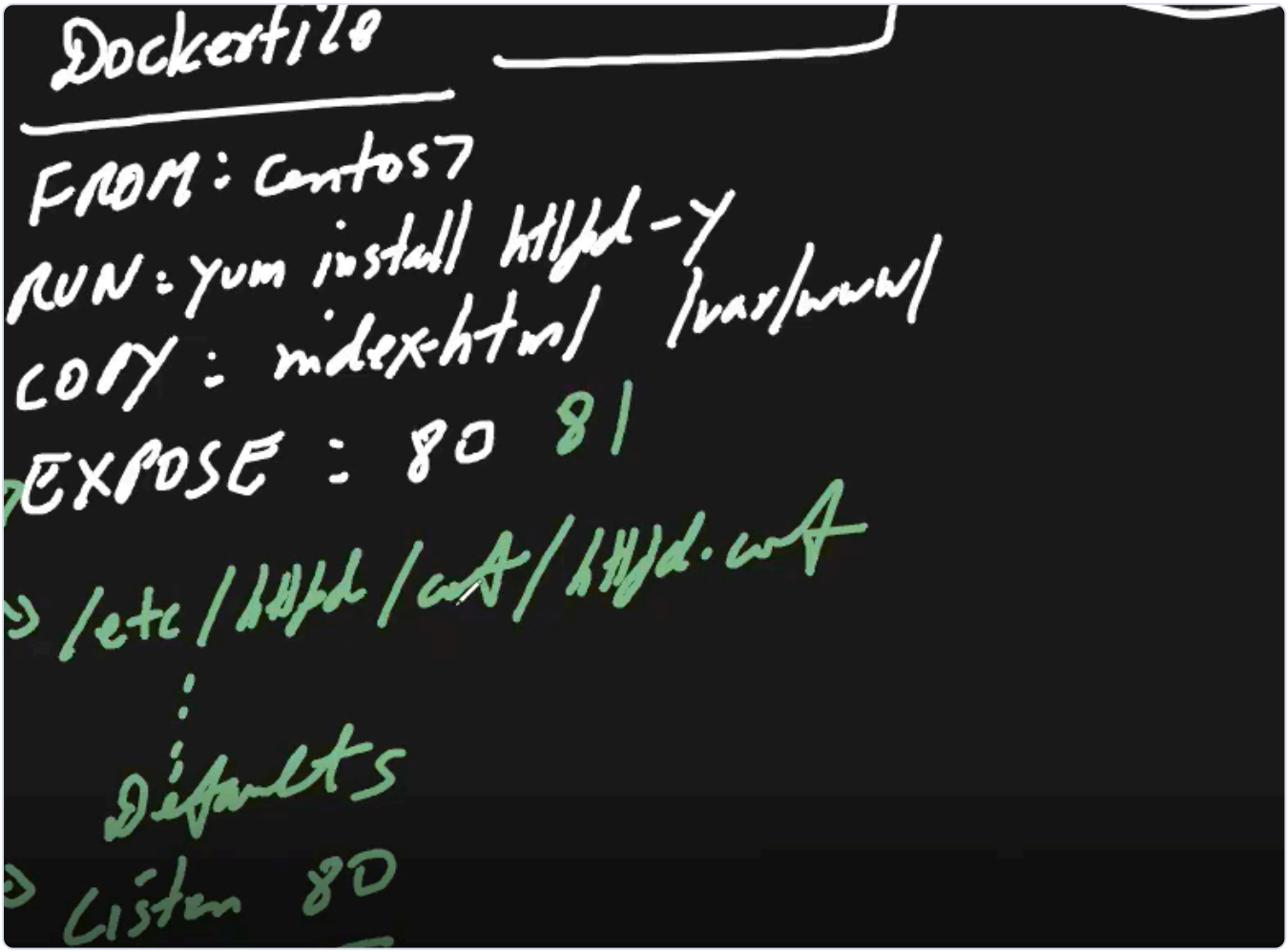

root@manager:~/weekends# kubectl exec -it secret /bin/bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use
kubectl exec [POD] -- [COMMAND] instead.
[root@secret /]# env
LANG=en_US.UTF-8
SECRET_PASSWORD=sunday
HOSTNAME=secret
HELLO_DEPLOY_SVC_PORT_80_TCP_ADDR=10.98.9.89
HELLO_DEPLOY_SVC_SERVICE_HOST=10.98.9.89
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
SECRET_USERNAME=alok
KUBERNETES_PORT=tcp://10.96.0.1:443
PWD=/
HOME=/root

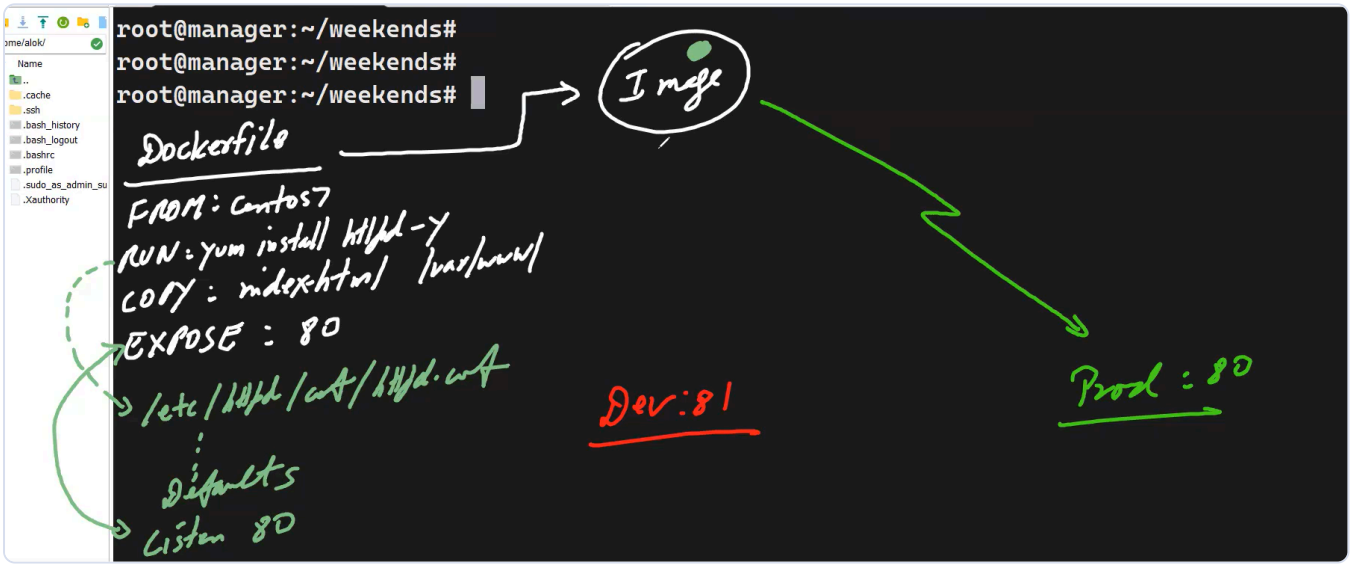

Config Map

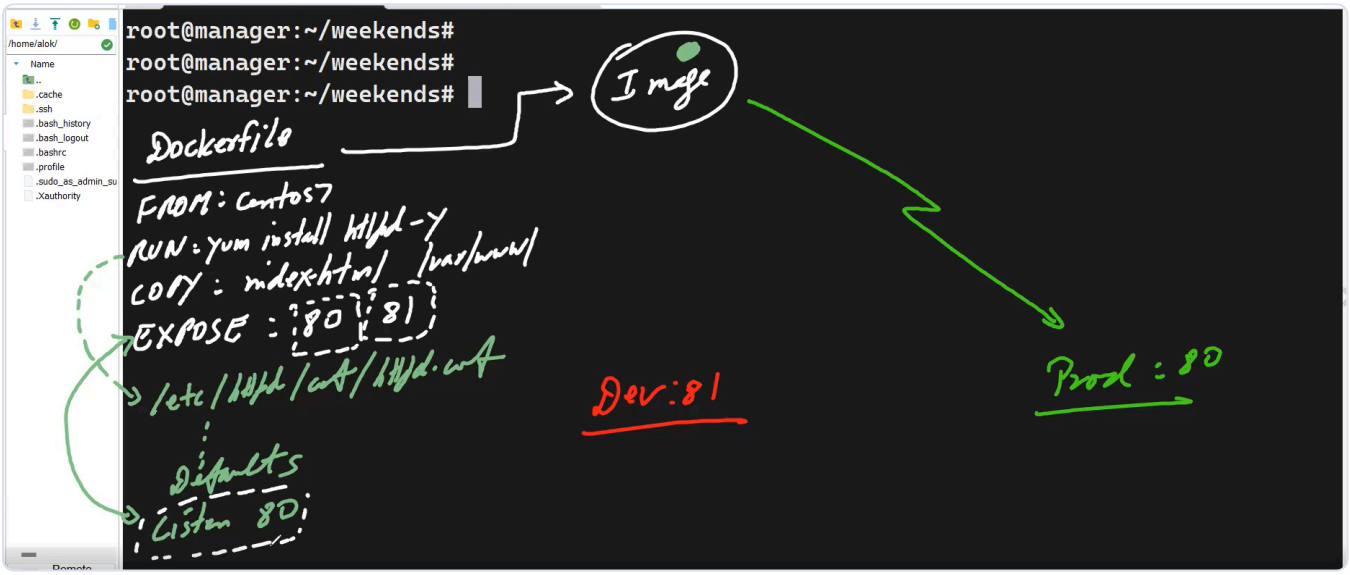

t@manager:~/weekends#
t@manager:~/weekends#
t@manager:~/weekends#
Config Map
Image
CM

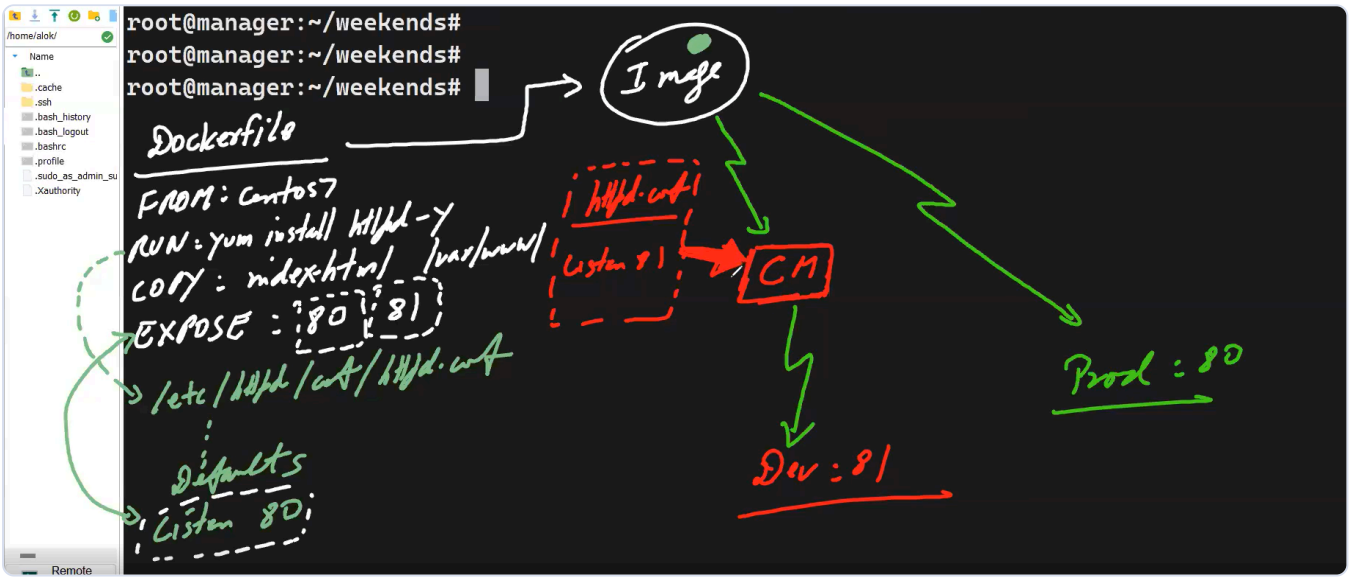

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Image
Dockerfile
FROM: centos7
RUN: yum install httpd -y
COPY: index.html /var/www/
EXPOSE: 80

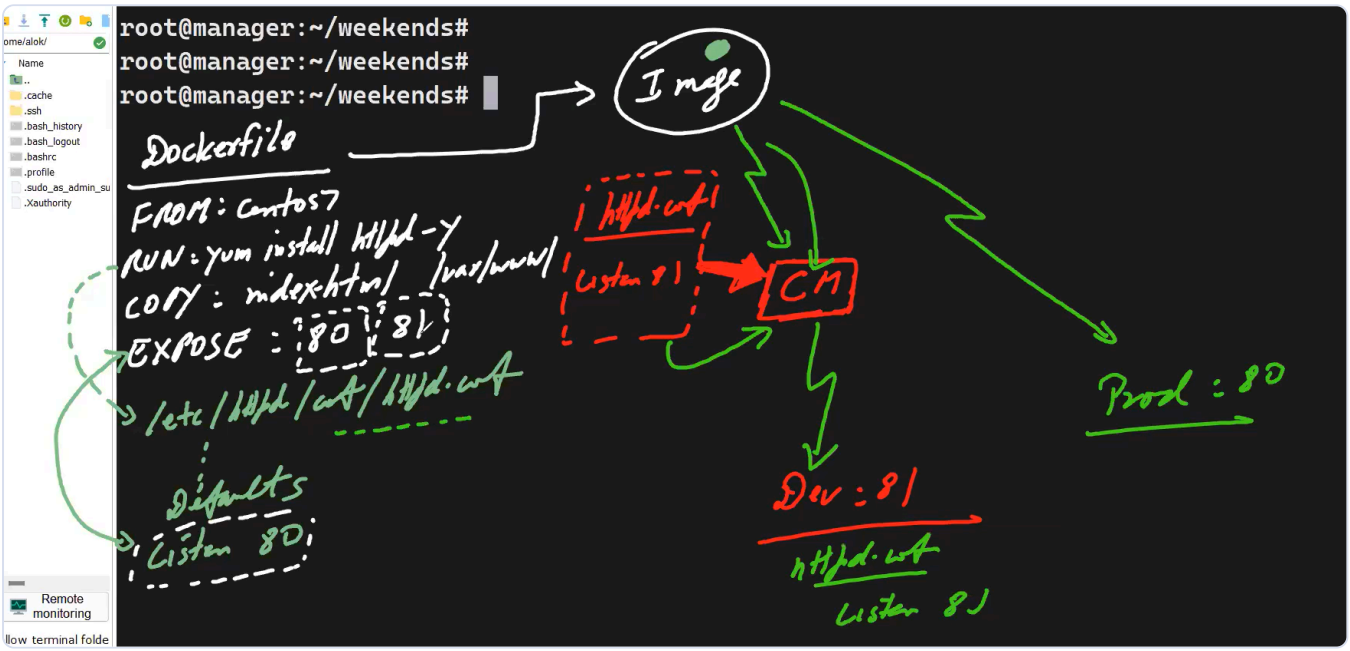

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Image
Dockerfile
FROM: Centos7
RUN: yum install httpd -y
COPY: index.html /var/www/
EXPOSE: 80
/etc/httpd/conf/httpd.conf
Defaults
Listen 80

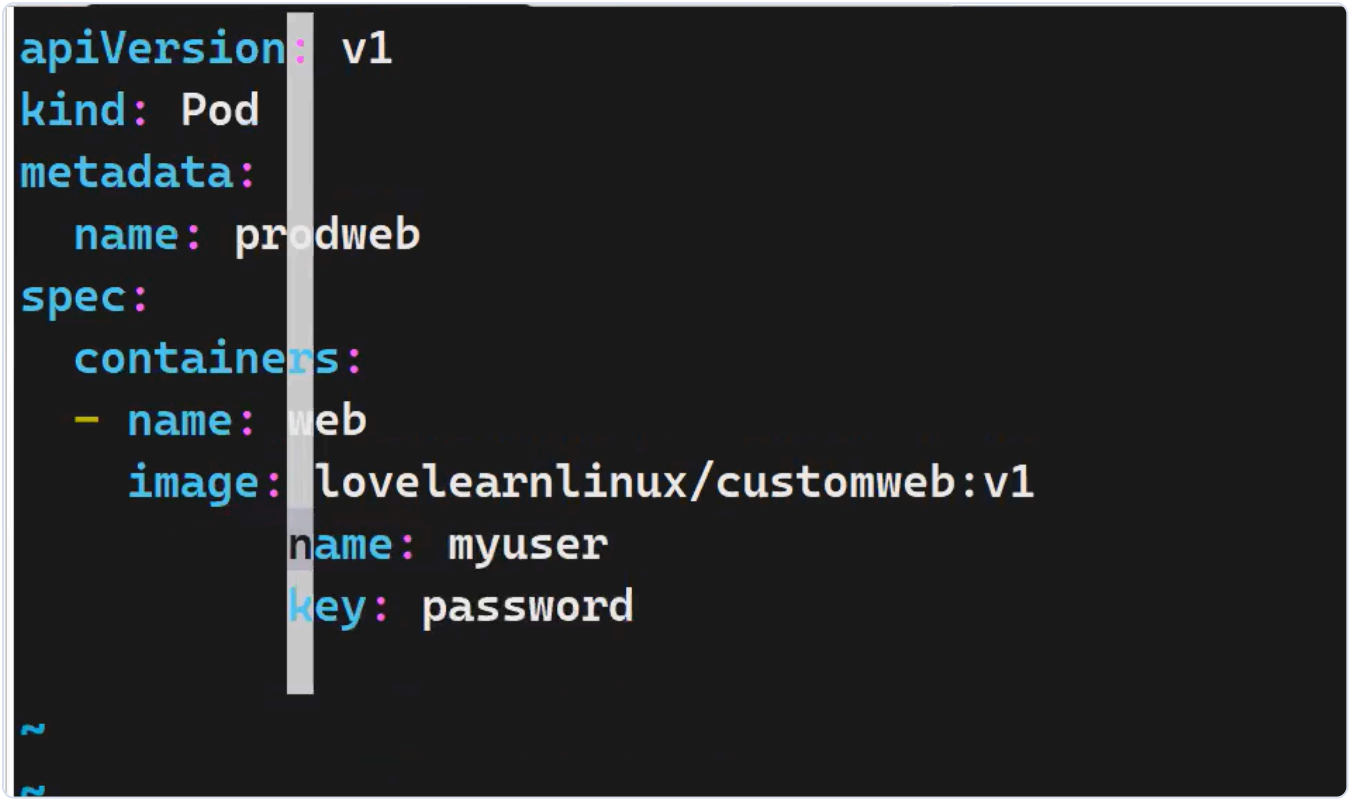

Dockerfile
FROM : Centos7
RUN : yum install httpd -y
COPY : index.html /var/www/
EXPOSE : 80 81
/etc/httpd/conf/httpd.conf
Defaults
Listen 80

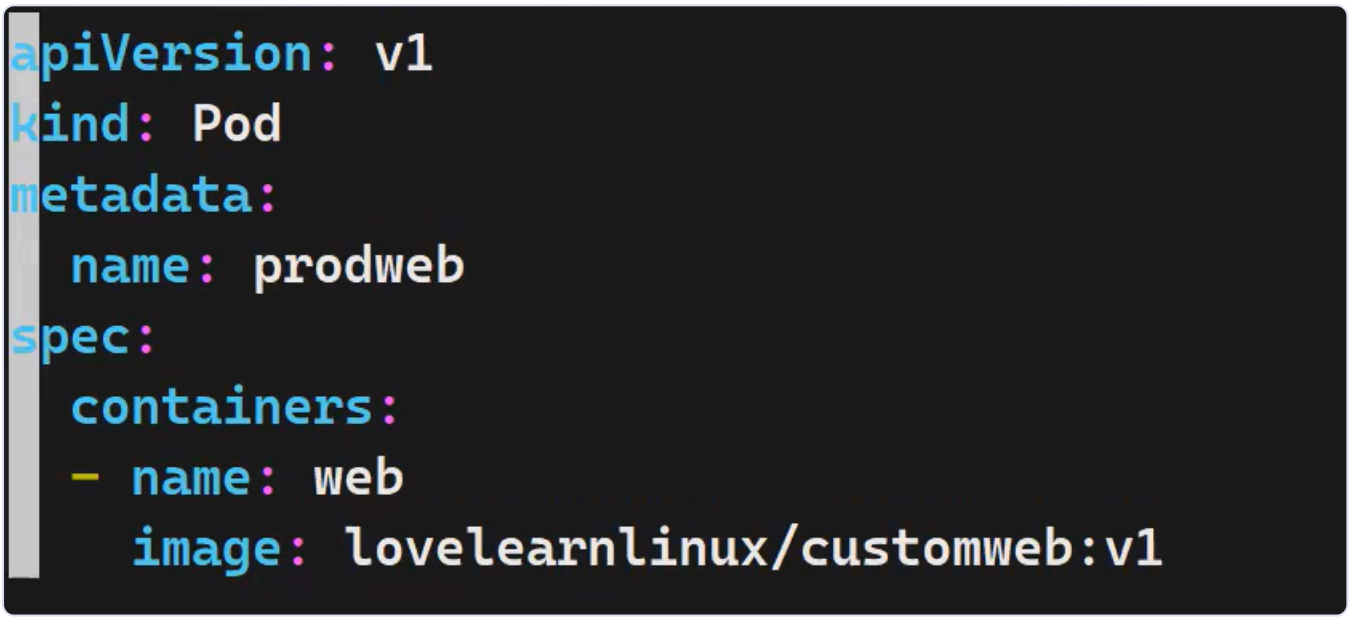

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Image
Dockerfile
FROM: centos7
RUN: yum install httpd -y
COPY: index.html /var/www/
EXPOSE: 80
/etc/httpd/conf/httpd.conf
Defaults
Listen 80
Dev: 81
Prod: 80

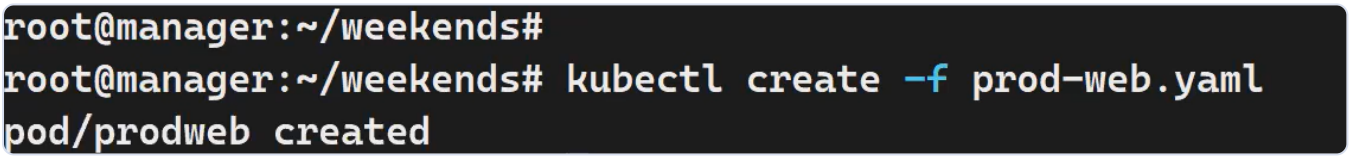

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Dockerfile
FROM: centos7
RUN: yum install httpd -y
COPY: index.html /var/www/
EXPOSE: 80 81
/etc/httpd/conf/httpd.conf
Defaults
Listen 80
Image
Dev: 81
Prod: 80

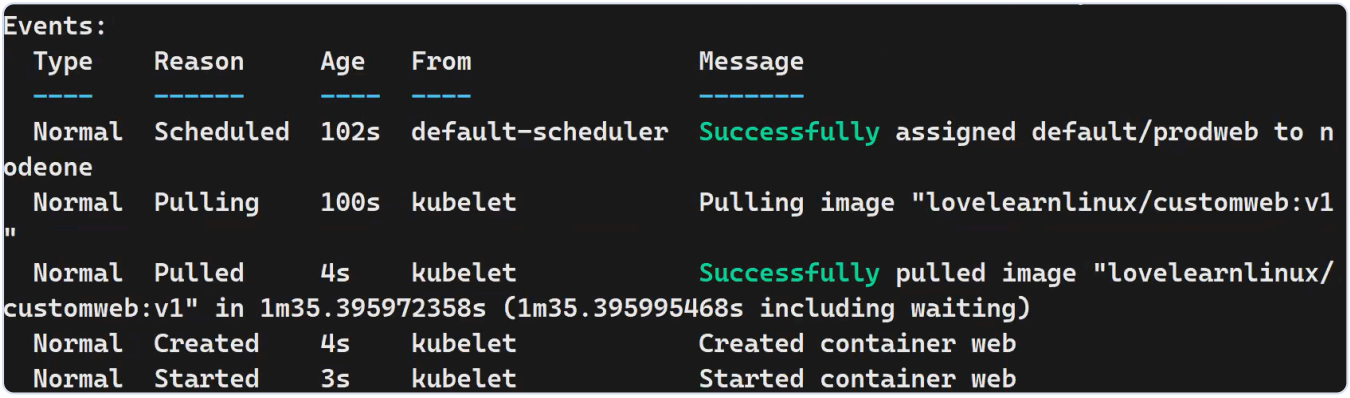

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Image
Dockerfile
FROM : centos7
RUN : yum install httpd -y
COPY : index.html /var/www/
EXPOSE : 80 81
/etc/httpd/conf/httpd.conf
Defaults
Listen 80
httpd.conf
Listen 81
CM
Prod : 80
Dev : 81
/home/alok/
Name
..
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_su
.Xauthority
Remote

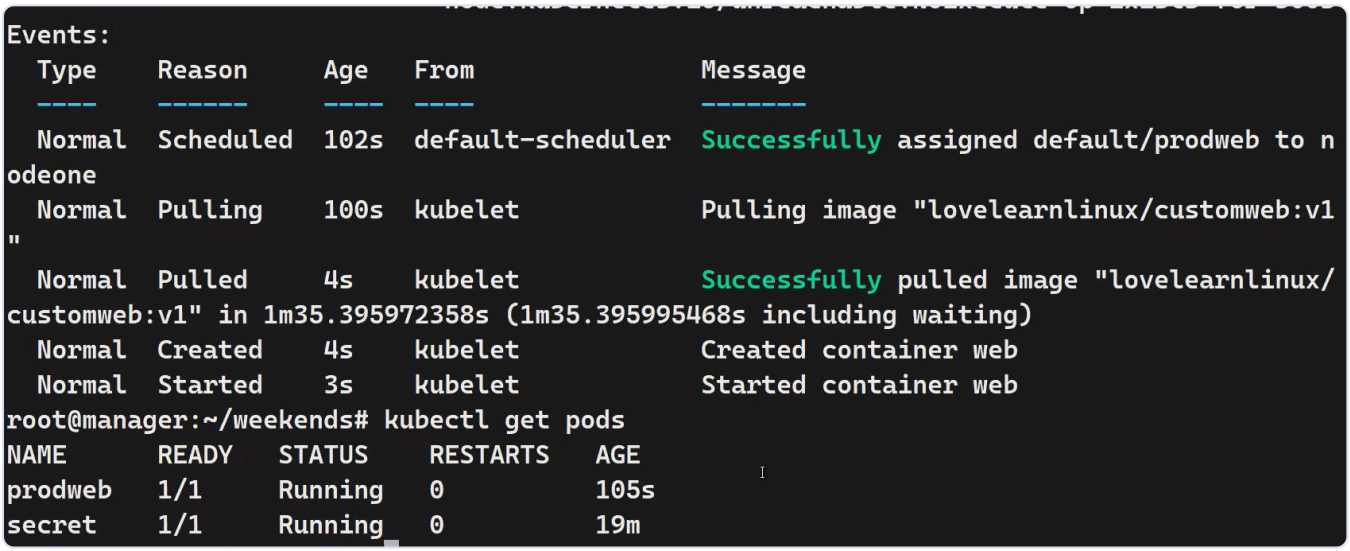

root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends#
Image
Dockerfile
FROM: centos7
RUN: yum install httpd -y
COPY: index.html /var/www/
EXPOSE: 80 81
/etc/httpd/conf/httpd.conf
Defaults
Listen 80
httpd.conf
Listen 81
CM
Prod: 80
Dev: 81
httpd.conf
Listen 81
Name
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_su
.Xauthority
Remote monitoring

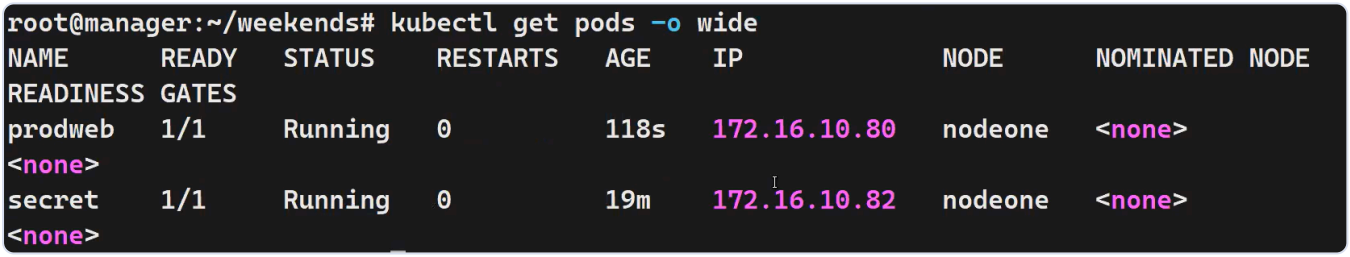

apiVersion: v1
kind: Pod
metadata:
name: prodweb
spec:
containers:
- name: web
image: lovelearnlinux/customweb:v1
name: myuser
key: password
~
~


apiVersion: v1
kind: Pod
metadata:
name: prodweb
spec:
containers:
- name: web
image: lovelearnlinux/customweb:v1


root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f prod-web.yaml
pod/prodweb created

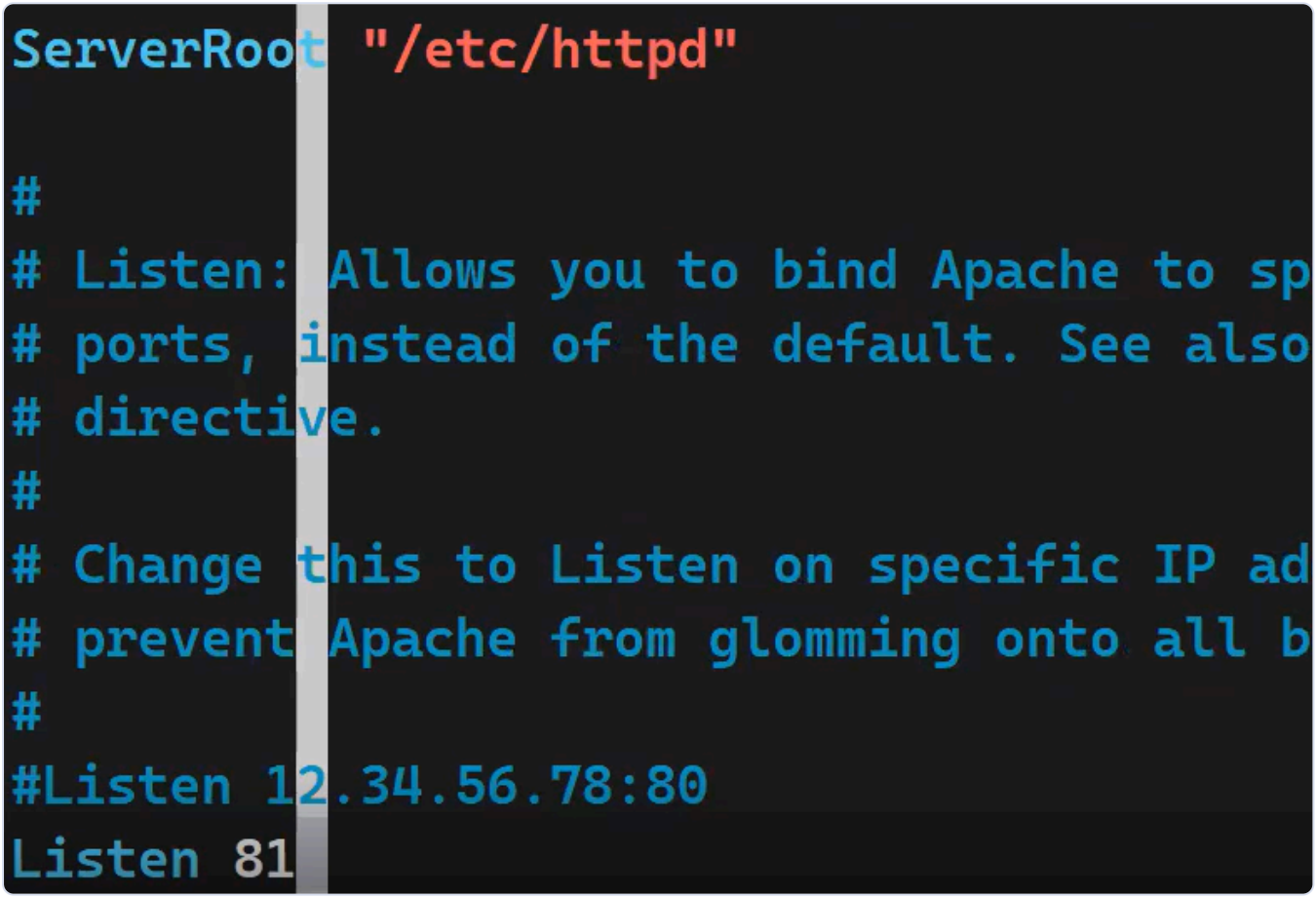

Events:
Type Reason Age From Message
Normal Scheduled 102s default-scheduler Successfully assigned default/prodweb to nodeone
Normal Pulling 100s kubelet Pulling image "lovelearnlinux/customweb:v1"
Normal Pulled 4s kubelet Successfully pulled image "lovelearnlinux/customweb:v1" in 1m35.395972358s (1m35.395995468s including waiting)
Normal Created 4s kubelet Created container web
Normal Started 3s kubelet Started container web


Events:
Type Reason Age From Message
Normal Scheduled 102s default-scheduler Successfully assigned default/prodweb to nodeone
Normal Pulling 100s kubelet Pulling image "lovelearnlinux/customweb:v1"
Normal Pulled 4s kubelet Successfully pulled image "lovelearnlinux/customweb:v1" in 1m35.395972358s (1m35.395995468s including waiting)
Normal Created 4s kubelet Created container web
Normal Started 3s kubelet Started container web
root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
prodweb 1/1 Running 0 105s
secret 1/1 Running 0 19m

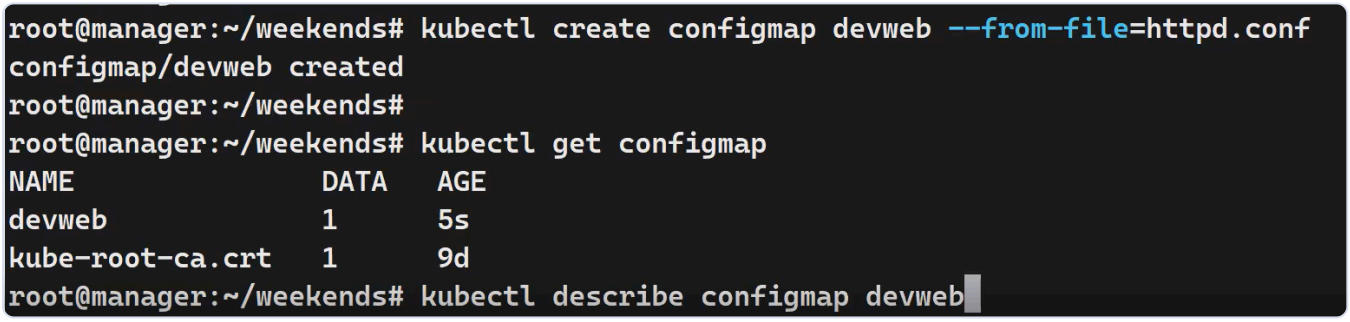

root@manager:~/weekends# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE
READINESS GATES
prodweb 1/1 Running 0 118s 172.16.10.80 nodeone <none>
<none>
secret 1/1 Running 0 19m 172.16.10.82 nodeone <none>
<none>

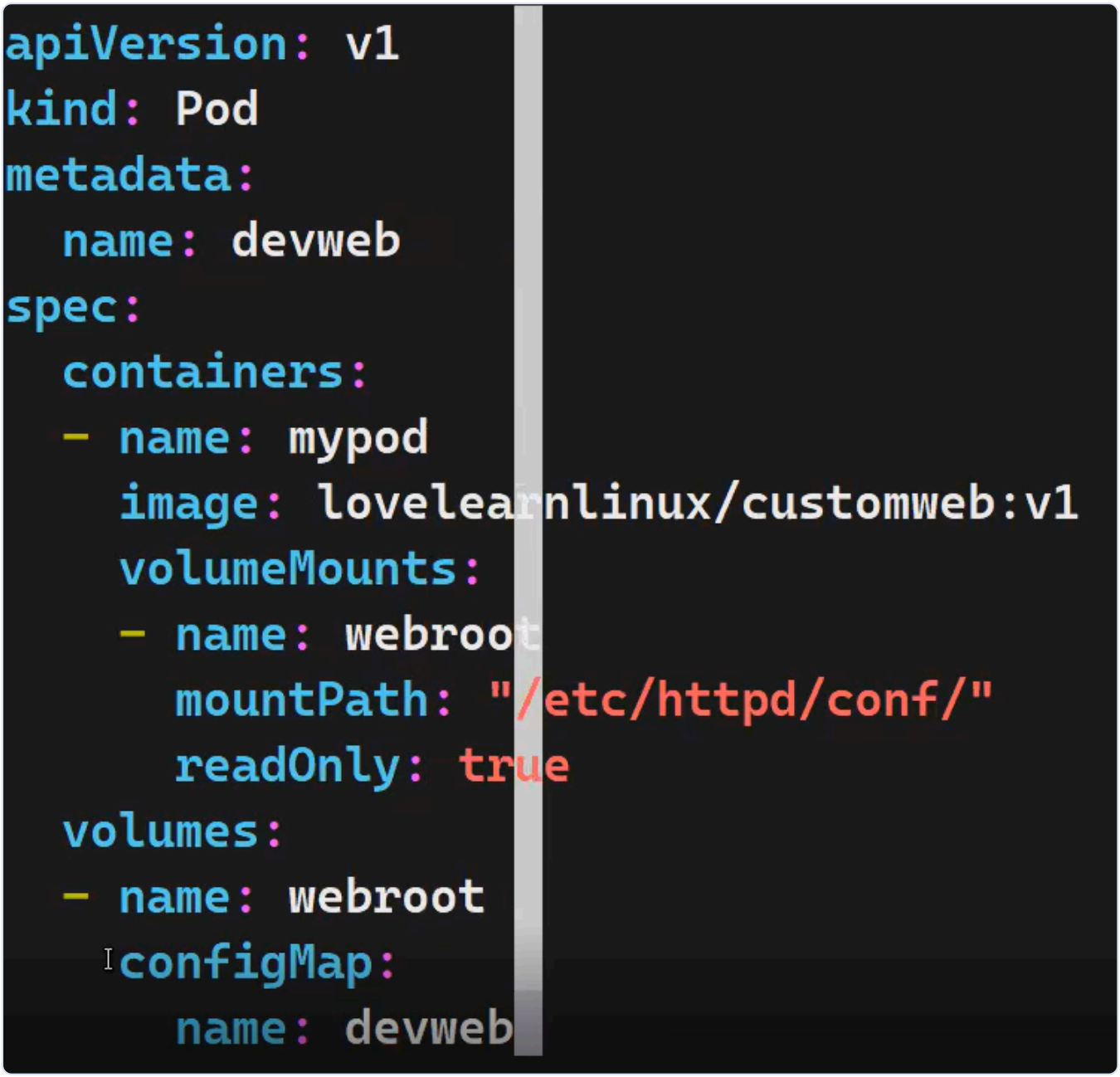

root@manager:~/weekends# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE
READINESS GATES
prodweb 1/1 Running 0 118s 172.16.10.80 nodeone <none>
<none>
secret 1/1 Running 0 19m 172.16.10.82 nodeone <none>
<none>
root@manager:~/weekends# curl http://172.16.10.80
this is a custom webserver
which can run on port 80 and 81

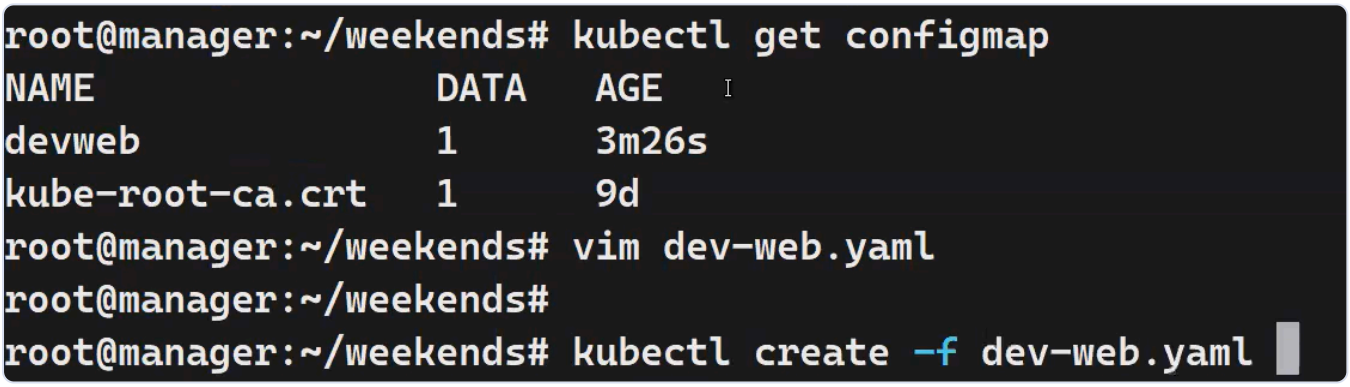

cat /etc/httpd/conf/httpd.conf

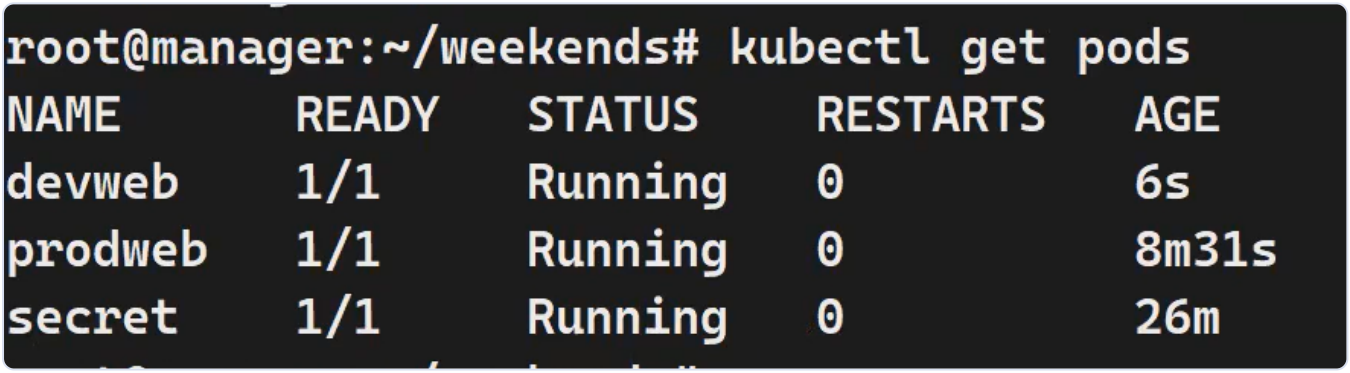

ServerRoot "/etc/httpd"
#
Listen: Allows you to bind Apache to sp
ports, instead of the default. See also
directive.
#
Change this to Listen on specific IP ad
prevent Apache from glomming onto all b
#
#Listen 12.34.56.78:80
Listen 81

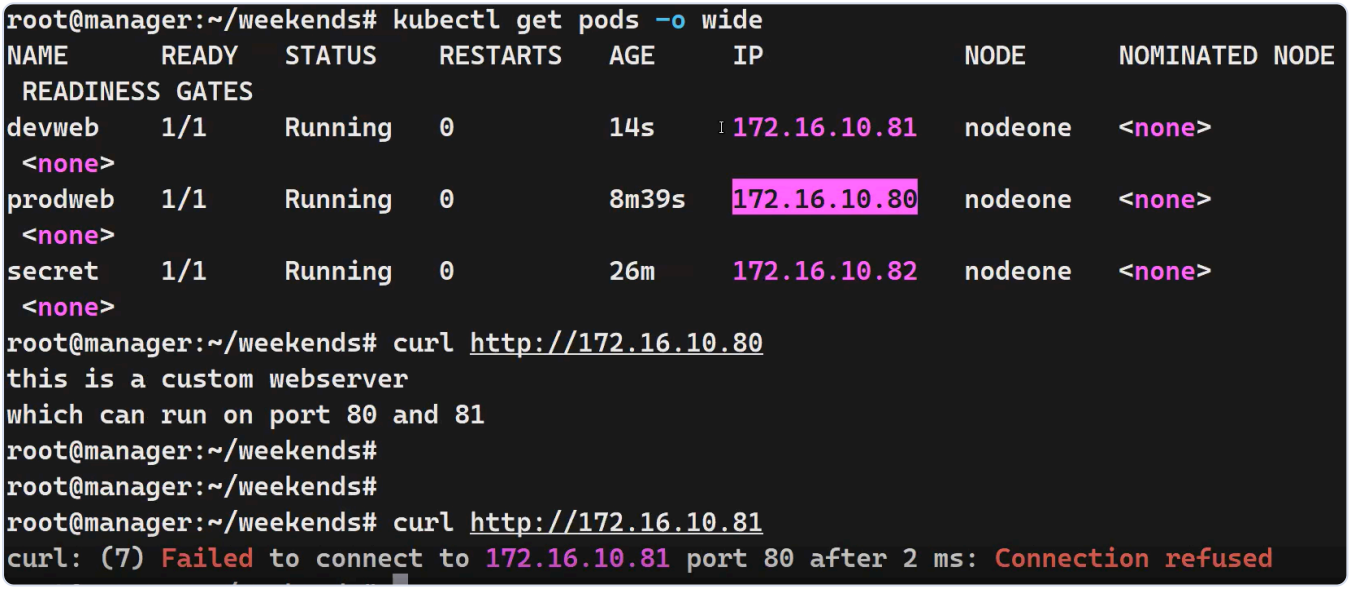
```
kubectl create configmap devweb --from-file httpd.conf
```


root@manager:~/weekends# kubectl create configmap devweb --from-file=httpd.conf
configmap/devweb created
root@manager:~/weekends#
root@manager:~/weekends# kubectl get configmap
NAME   DATA   AGE
devweb   1   5s
kube-root-ca.crt   1   9d
root@manager:~/weekends# kubectl describe configmap devweb


apiVersion: v1
kind: Pod
metadata:
name: devweb
spec:
containers:
- name: mypod
image: lovelearnlinux/customweb:v1
volumeMounts:
- name: webroot
mountPath: "/etc/httpd/conf/"
readOnly: true
volumes:
- name: webroot
configMap:
name: devweb

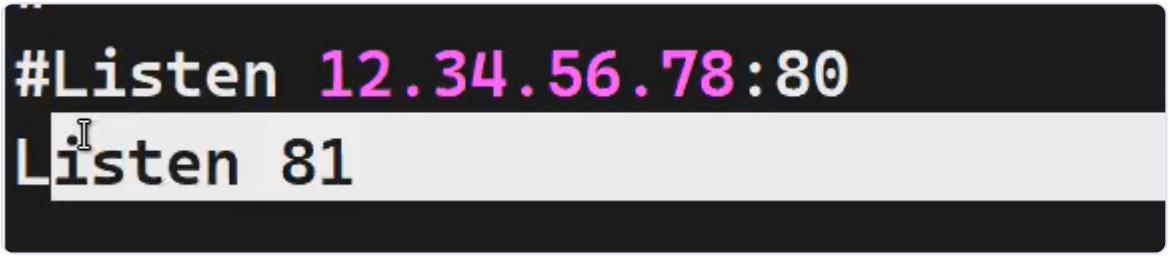

root@manager:~/weekends# kubectl get configmap
NAME DATA AGE
devweb 1 3m26s
kube-root-ca.crt 1 9d
root@manager:~/weekends# vim dev-web.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f dev-web.yaml


root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
devweb 1/1 Running 0 6s
prodweb 1/1 Running 0 8m31s
secret 1/1 Running 0 26m


root@manager:~/weekends# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE
READINESS GATES
devweb 1/1 Running 0 14s 172.16.10.81 nodeone <none>
<none>
prodweb 1/1 Running 0 8m39s 172.16.10.80 nodeone <none>
<none>
secret 1/1 Running 0 26m 172.16.10.82 nodeone <none>
<none>
root@manager:~/weekends# curl http://172.16.10.80
this is a custom webserver
which can run on port 80 and 81
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# curl http://172.16.10.81
curl: (7) Failed to connect to 172.16.10.81 port 80 after 2 ms: Connection refused

```
NAME      READY   STATUS    RESTARTS   AGE     IP              NODE      NOMINATED NODE
 READINESS GATES
devweb    1/1     Running   0          14s     172.16.10.81    nodeone   <none>
 <none>
prodweb   1/1     Running   0          8m39s   172.16.10.80    nodeone   <none>
 <none>
secret    1/1     Running   0          26m     172.16.10.82    nodeone   <none>
 <none>
root@manager:~/weekends# curl http://172.16.10.80
this is a custom webserver
which can run on port 80 and 81
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# curl http://172.16.10.81
curl: (7) Failed to connect to 172.16.10.81 port 80 after 2 ms: Connection refused
root@manager:~/weekends# curl http://172.16.10.81:81
this is a custom webserver
which can run on port 80 and 81
```


root@manager:~/weekends# kubectl exec -it devweb /bin/bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use
kubectl exec [POD] -- [COMMAND] instead.
[root@devweb /]# cat /etc/httpd/conf/httpd.conf | less


#Listen 12.34.56.78:80
Listen 81